

Enterprise SOC Home Lab Using Wazuh SIEM

Comprehensive Project Report

Abstract

This project documents the end-to-end deployment of a Security Operations Center (SOC) home lab using Ubuntu Server, Wazuh SIEM, Kali Linux, VirtualBox networking, custom detection rules, threat hunting, and security monitoring.

1. Introduction

A SIEM platform centralizes logs, alerts, and security events. Wazuh was selected because it combines log collection, threat detection, file integrity monitoring, vulnerability assessment, and threat hunting.

2. Objectives

Deploy Wazuh, connect endpoints, perform threat hunting, create custom detections, investigate alerts, and simulate SOC analyst workflows.

3. Lab Architecture

Ubuntu Server hosted Wazuh Manager, Dashboard, and Indexer. Kali Linux acted as attacker and monitored endpoint. Internal networking was configured using VirtualBox.

4. Network Configuration

VirtualBox Adapter 1 was configured as NAT for internet access. Adapter 2 was configured as Internal Network named SOC-LAB for isolated communication.

Commands Used

ip a - display interfaces and IP addresses.
ifconfig -a - display network interfaces.
ping - verify connectivity between systems.

Static IP Assignment

```
sudo ip addr add 192.168.56.10/24 dev enp0s8
```

Assigned static IP to Ubuntu.

```
sudo ip addr add 192.168.56.20/24 dev eth1
```

Assigned static IP to Kali.

Connectivity Verification

Systems successfully communicated using ICMP ping. This validated network design and endpoint reachability.

5. Wazuh Installation

The official all-in-one installer was used to deploy Wazuh Manager, Dashboard and Indexer.

Installation Commands

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh  
chmod +x wazuh-install.sh  
sudo ./wazuh-install.sh -a
```

Hardware Challenge

The installer reported insufficient memory. Ubuntu VM RAM was increased and installation continued successfully.

Storage Challenge

Indexer entered read-only mode due to flood-stage watermark. Investigation revealed the filesystem was nearly full.

Storage Investigation Commands

```
df -h  
lsblk
```

These commands were used to analyze disk usage and LVM allocation.

LVM Expansion

```
sudo lvextend -l +100%FREE /dev/ubuntu-vg/ubuntu-lv  
sudo resize2fs /dev/ubuntu-vg/ubuntu-lv
```

Expanded filesystem from approximately 24GB to nearly the full virtual disk size.

6. Dashboard Access

The Wazuh dashboard was accessed through a browser using HTTPS. Internal networking architecture was analyzed to troubleshoot access issues.

7. Agent Deployment

Kali Linux was enrolled as a Wazuh agent and successfully connected to the manager.

Agent Installation

```
wget  
sudo WAZUH_MANAGER='192.168.56.10' WAZUH_AGENT_NAME='kali-attacker' dpkg -i  
wazuh-agent.deb  
sudo systemctl enable wazuh-agent  
sudo systemctl start wazuh-agent
```

Agent Verification

The agent appeared as Active in the Wazuh Dashboard and started sending events.

8. Threat Hunting

Threat Hunting was performed using the Discover interface. Logs from authentication, sudo activity, and rootcheck modules were analyzed.

Observed Alerts

PAM login session opened.
PAM login session closed.
Successful sudo to ROOT executed.
Rootcheck findings.

9. Custom Detection Engineering

A custom rule was developed to detect reconnaissance activity.

Custom Rule

```
nmap  
Nmap Reconnaissance Detected  
recon,custom,
```

Rule Deployment Commands

```
sudo nano /var/ossec/etc/rules/local_rules.xml  
sudo /var/ossec/bin/wazuh-analysisd -t  
sudo systemctl restart wazuh-manager
```

Rule Testing

```
echo 'nmap scan detected' | logger  
The generated event triggered the custom rule and produced a high-severity alert.
```

10. File Integrity Monitoring

File creation, modification, permission changes, and deletion were tested.

FIM Commands

```
sudo touch /etc/resume-test.txt  
echo 'modified by pavan' | sudo tee -a /etc/resume-test.txt  
sudo chmod 777 /etc/resume-test.txt  
sudo rm /etc/resume-test.txt
```

11. Privilege Escalation Monitoring

sudo whoami and sudo operations generated alerts. Wazuh detected successful elevation events and mapped them to predefined rules.

12. Skills Demonstrated

SIEM Deployment, Threat Hunting, Detection Engineering, Linux Administration, Networking, Log Analysis, Security Monitoring, Alert Investigation, Virtualization.

13. Challenges and Troubleshooting

Network adapter configuration, dashboard accessibility, XML rule validation, disk capacity management, and service troubleshooting were resolved during deployment.

14. Future Enhancements

Deploy Metasploitable monitoring, integrate Suricata IDS, enable Active Response, perform SSH brute-force detection, configure Vulnerability Detection, and build MITRE ATT&CK; dashboards.

15. Real-World Relevance

This project simulates SOC analyst responsibilities including alert triage, detection engineering, log analysis, endpoint monitoring, and incident investigation.

Conclusion

The project successfully established a functional SOC environment using Wazuh. The lab provided hands-on experience with monitoring, detection, threat hunting, and SIEM administration.